Decentralized Service Mesh Resilience and Dynamic Autoscaling in Cloud-Native Architectures

Authors: Dr. Elena Rostova, Marcus Sterling, Alex Kumar
Affiliation: Cloud-Native Systems Institute & Stanford AI Lab
Published: IEEE Transactions on Cloud Computing, 2026
DOI: 10.1109/TCC.2026.8841920

ABSTRACT
Modern enterprise cloud platforms increasingly migrate monolithic core workloads into microservices deployed across containe

KEYWORDS
Cloud-Native; Microservices; Service Mesh; Kubernetes; Fault Tolerance; API Gateway; Chaos Engineering; Distributed Systen

1. INTRODUCTION & RESEARCH PROBLEM
The proliferation of microservices topologies introduces complex dependency graphs where single-point microservice failures ca

2. METHODOLOGY & EXPERIMENTAL SETUP

We engineered a synthetic microservices benchmark consisting of 24 polyglot services deployed across a 64-node Kubernetes

- Simulated packet loss ranging from 5% to 35%
- Latency injection spikes up to 2500ms
- Sudden volumetric request surges up to 100,000 requests per second

Telemetry was collected at the API gateway layer and sidecar proxies using OpenTelemetry standards.

3. MAJOR FINDINGS & RESULTS

Our quantitative analysis yielded three primary results:

1. Adaptive circuit breakers decoupled downstream bottlenecks, preventing thread pool exhaustion in upstream API gateways.
2. Distributed token-bucket rate limiting at ingress proxies successfully neutralized layer-7 denial-of-service surges.
3. Mean Time to Recovery (MTTR) under chaotic node termination improved from 142 seconds to 18 seconds using automated

4. LIMITATIONS & RESEARCH GAPS

Current service mesh implementations exhibit non-trivial memory footprints on ultra-dense edge nodes. Additionally, cross-clust

5. FUTURE DIRECTIONS

Future research will investigate embedding lightweight reinforcement learning models directly into Envoy-compatible proxy filter